

## VIANA DO CASTELO, Portugal

WMO: 085430

Lat: 41.695N

Lon: 8.830W

Elev: 18

StdP: 101.11

Time Zone: 0.00 (EUW)

Period: 94-06

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 0.1        | 1.4 | -3.1                          | 2.9 | 6.2  | -1.3 | 3.4 | 5.5  | 8.0                   | 13.5 | 7.1 | 11.9 | 0.3                   | 50   | 0.388 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.1                   | 33.5            | 21.2 | 31.3 | 20.7 | 29.1 | 20.0 | 22.2                | 30.7 | 21.4 | 29.0 | 20.6 | 27.1 | 2.1                  | 10   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.6                            | 14.4 | 24.1 | 19.0 | 13.8 | 23.4 | 18.4 | 13.3 | 22.6 | 65.0          | 30.5 | 61.8 | 29.2 | 59.1 | 27.1 | 26.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 6.9           | 5.8  | 5.0 | -2.1                       | 36.8 | 1.7                | 1.3 | -3.3                                               | 37.7 | -4.3       | 38.5 | -5.3       | 39.2 | -6.5       | 40.2 |    |    |
| (5)               |      |     | -2.5                       | 24.4 | 1.9                | 1.4 | -3.9                                               | 25.4 | -5.0       | 26.2 | -6.0       | 27.0 | -7.4       | 28.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 15.2   | 10.0  | 10.4  | 12.9  | 13.5  | 16.0  | 19.8  | 20.7  | 20.8  | 19.4  | 16.2  | 12.4  | 10.5  |  |  |
|                                                                        | DBStd       | 4.78   | 2.65  | 2.46  | 2.60  | 2.43  | 2.87  | 3.31  | 2.75  | 2.50  | 2.28  | 2.41  | 2.63  | 2.62  |  |  |
|                                                                        | HDD10.0     | 99     | 34    | 23    | 6     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 7     | 27    |  |  |
|                                                                        | HDD18.3     | 1420   | 259   | 223   | 169   | 147   | 86    | 19    | 6     | 3     | 13    | 73    | 177   | 244   |  |  |
|                                                                        | CDD10.0     | 2009   | 33    | 34    | 95    | 106   | 186   | 294   | 331   | 333   | 282   | 193   | 80    | 41    |  |  |
|                                                                        | CDD18.3     | 287    | 0     | 0     | 1     | 2     | 13    | 64    | 78    | 78    | 44    | 8     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 2792   | 0     | 0     | 29    | 46    | 172   | 720   | 756   | 617   | 388   | 65    | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 1074   | 0     | 0     | 3     | 10    | 56    | 311   | 314   | 239   | 132   | 10    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 1.7    | 1.6   | 1.6   | 1.8   | 2.2   | 1.9   | 1.7   | 1.8   | 1.5   | 1.3   | 1.7   | 1.5   | 1.4   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1422   | 190   | 170   | 110   | 112   | 102   | 53    | 24    | 27    | 91    | 156   | 176   | 211   |  |  |
|                                                                        | PrecMax     | 1906   | 432   | 482   | 346   | 349   | 255   | 189   | 81    | 97    | 366   | 533   | 492   | 583   |  |  |
|                                                                        | PrecMin     | 858    | 18    | 5     | 0     | 20    | 4     | 0     | 0     | 0     | 1     | 7     | 0     | 28    |  |  |
|                                                                        | PrecStd     | 268    | 106   | 109   | 78    | 73    | 61    | 36    | 23    | 27    | 79    | 103   | 112   | 124   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 19.1  | 21.1  | 26.8  | 28.2  | 31.7  | 35.0  | 35.1  | 36.4  | 32.8  | 28.0  | 22.0  | 18.2  |  |  |
|                                                                        |             | MCWB   | 12.8  | 14.2  | 16.3  | 17.4  | 20.3  | 21.2  | 21.3  | 21.4  | 20.3  | 19.0  | 15.5  | 13.9  |  |  |
|                                                                        | 2%          | DB     | 16.6  | 18.8  | 23.1  | 23.9  | 27.7  | 32.8  | 32.6  | 32.5  | 30.4  | 25.0  | 19.5  | 16.5  |  |  |
|                                                                        |             | MCWB   | 12.4  | 12.9  | 14.4  | 15.5  | 18.0  | 21.1  | 21.3  | 21.3  | 20.2  | 17.6  | 15.0  | 13.7  |  |  |
|                                                                        | 5%          | DB     | 15.4  | 16.9  | 20.8  | 21.1  | 24.7  | 30.4  | 30.4  | 29.0  | 27.7  | 22.7  | 18.1  | 15.6  |  |  |
|                                                                        |             | MCWB   | 12.3  | 12.0  | 13.9  | 14.4  | 17.4  | 20.0  | 20.9  | 20.4  | 19.1  | 17.3  | 14.7  | 13.5  |  |  |
|                                                                        | 10%         | DB     | 14.5  | 15.2  | 18.1  | 18.8  | 21.8  | 27.6  | 27.6  | 26.3  | 25.1  | 20.7  | 17.0  | 14.9  |  |  |
|                                                                        |             | MCWB   | 12.4  | 11.4  | 13.0  | 13.6  | 16.1  | 19.3  | 20.0  | 19.6  | 18.5  | 16.9  | 14.4  | 13.0  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 15.1  | 15.4  | 16.9  | 18.4  | 22.5  | 22.7  | 23.2  | 23.8  | 21.7  | 20.2  | 17.9  | 15.8  |  |  |
|                                                                        |             | MCDB   | 16.0  | 19.6  | 23.2  | 26.1  | 29.7  | 32.0  | 31.7  | 32.7  | 29.9  | 23.8  | 18.8  | 16.3  |  |  |
|                                                                        | 2%          | WB     | 14.1  | 14.2  | 15.6  | 16.3  | 19.2  | 21.8  | 22.0  | 22.0  | 20.7  | 19.2  | 16.7  | 15.1  |  |  |
|                                                                        |             | MCDB   | 14.7  | 16.7  | 19.8  | 22.0  | 26.0  | 31.2  | 30.5  | 29.2  | 27.3  | 22.4  | 18.0  | 15.7  |  |  |
|                                                                        | 5%          | WB     | 13.6  | 13.3  | 14.8  | 15.2  | 17.8  | 20.7  | 21.1  | 21.0  | 19.9  | 18.3  | 15.9  | 14.4  |  |  |
|                                                                        |             | MCDB   | 14.5  | 15.1  | 18.0  | 19.4  | 23.1  | 28.1  | 29.1  | 27.0  | 25.5  | 21.0  | 17.3  | 15.2  |  |  |
|                                                                        | 10%         | WB     | 12.9  | 12.4  | 14.2  | 14.3  | 16.6  | 19.7  | 20.1  | 20.2  | 19.2  | 17.5  | 14.9  | 13.5  |  |  |
|                                                                        |             | MCDB   | 14.4  | 14.4  | 17.0  | 17.3  | 20.6  | 25.9  | 26.6  | 25.3  | 23.8  | 19.9  | 16.4  | 14.6  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.6   | 10.1  | 10.1  | 9.3   | 9.7   | 11.1  | 11.1  | 11.0  | 11.0  | 9.1   | 8.8   | 8.4   |  |  |
|                                                                        |             | MCDBR  | 10.2  | 13.6  | 15.3  | 15.0  | 15.4  | 16.8  | 16.5  | 15.9  | 16.3  | 13.0  | 10.7  | 9.2   |  |  |
|                                                                        |             | MCWBR  | 6.8   | 8.3   | 8.2   | 8.0   | 7.8   | 7.5   | 7.5   | 7.2   | 7.3   | 7.1   | 6.7   | 6.5   |  |  |
|                                                                        | 5% WB       | MCDBR  | 6.0   | 9.6   | 11.4  | 12.4  | 13.0  | 15.2  | 14.9  | 13.5  | 12.8  | 9.8   | 7.1   | 5.4   |  |  |
|                                                                        |             | MCWBR  | 4.9   | 6.6   | 6.9   | 7.2   | 7.1   | 7.2   | 7.2   | 7.0   | 6.6   | 5.9   | 5.3   | 4.9   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.340 | 0.348 | 0.383 | 0.382 | 0.388 | 0.391 | 0.380 | 0.392 | 0.380 | 0.374 | 0.347 | 0.335 |  |  |
|                                                                        | taud        |        | 2.476 | 2.442 | 2.349 | 2.368 | 2.376 | 2.386 | 2.410 | 2.376 | 2.425 | 2.451 | 2.481 | 2.482 |  |  |
|                                                                        | Ebn at Noon |        | 805   | 856   | 861   | 889   | 888   | 883   | 889   | 868   | 854   | 814   | 789   | 776   |  |  |
|                                                                        | Edh at Noon |        | 76    | 91    | 112   | 117   | 119   | 119   | 115   | 115   | 102   | 89    | 75    | 70    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.76  | 2.78  | 3.90  | 5.21  | 6.32  | 6.88  | 6.96  | 6.29  | 4.91  | 3.14  | 2.03  | 1.59  |  |  |
|                                                                        | RadStd      |        | 0.29  | 0.39  | 0.58  | 0.56  | 0.54  | 0.46  | 0.39  | 0.37  | 0.34  | 0.38  | 0.34  | 0.24  |  |  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (15 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | +36     |

Nomenclature: See separate page